

Model

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

Locations are connected by a transportation network (S, E) , where $E \subseteq S \times S$ is a set of directed edges (links). Transportation investment $I_{k\ell}$ is chosen at the *edge* level, for each $(k, \ell) \in E$ — not directly for residence-workplace pairs, which need not be adjacent in the network. The utility cost of traveling through edge (k, ℓ) is

$$d_{k\ell} = d_{k\ell}(I_{k\ell}, n_{k\ell}),$$

where $n_{k\ell}$ is the equilibrium commuter flow on that edge. Investment reduces edge-level travel cost and congestion raises it:

$$\frac{\partial d_{k\ell}}{\partial I_{k\ell}} \leq 0, \quad \frac{\partial d_{k\ell}}{\partial n_{k\ell}} \geq 0.$$

Following Fajgelbaum and Schaal, 2020, I parametrize the edge-level cost as

$$d_{k\ell} = \delta_{k\ell}^\tau \frac{n_{k\ell}^\rho}{I_{k\ell}^\gamma},$$

where $\delta_{k\ell}^\tau > 0$ captures geographic features of edge (k, ℓ) and $\rho, \gamma > 0$.

Each residence-workplace pair (i, j) is connected by an exogenously known route $r_{ij} \subseteq E$ — a sequence of edges linking i to j , taken from GIS shortest-path data for roads or the observed line/transfer sequence for transit. The route is fixed: households do not choose among alternative paths between i and j , though they do choose (i, j) itself. Allowing endogenous multi-route choice, as in Allen and Arkolakis (2022) and Bordeu (2026), is a

natural extension, deferred here for tractability. The generalized commuting cost between i and j accumulates over the route,

$$\tau_{ij} = \prod_{(k,\ell) \in r_{ij}} d_{k\ell}(I_{k\ell}, n_{k\ell}),$$

and edge-level traffic aggregates every commuter flow whose route uses that edge,

$$n_{k\ell} = \sum_{(i,j) : (k,\ell) \in r_{ij}} N\pi_{ij}. \quad (1)$$

Because τ_{ij} is built from the routes' constituent edges rather than treated as a primitive of (i, j) directly, investment on a single edge affects every residence-workplace pair whose route passes through it, not only pairs adjacent to that edge — the network is the source of investment spillovers in this model (see the Transportation Authority subsection).

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i and a workplace j to maximize utility. Preferences over the numaire good Q_{ij} and residential floor space h_{ij}^R are Cobb-Douglas:

$$u_{ij}(\omega) = \frac{B_i}{\tau_{ij}} \left(\frac{Q_{ij}}{\alpha} \right)^\alpha \left(\frac{h_{ij}^R}{1-\alpha} \right)^{1-\alpha} v_{ij}(\omega),$$

where $B_i > 0$ is a residential amenity at location i , $\alpha \in (0, 1)$ is the expenditure share on the composite good, and $v_{ij}(\omega)$ is an idiosyncratic preference shock drawn independently for each (i, j) pair.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j - \phi(N_i, L_i),$$

where q_i^R is the residential rent at location i , w_j is the wage at workplace j . Optimal demands are $Q_{ij}^* = \alpha Y_{ij}$ and $h_{ij}^{R*} = (1 - \alpha) Y_{ij} / q_i^R$. Substituting into the utility function yields the deterministic component of indirect utility:

$$\tilde{V}_{ij} = \frac{B_i (w_j - \phi(N_i, L_i))}{\tau_{ij} (q_i^R)^{1-\alpha}}.$$

The idiosyncratic shock $v_{ij}(\omega)$ is drawn from a Fréchet distribution with $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, where $T_i > 0$ and $E_j > 0$ are location-specific scale parameters and $\epsilon > 1$ governs the dispersion of idiosyncratic tastes. By the Fréchet aggregation property, the probability that a household chooses residence i and workplace j is

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\Phi}, \quad \Phi \equiv \sum_{i'} \Phi_{i'}, \quad \Phi_i \equiv \sum_j T_i E_j \tilde{V}_{ij}^\epsilon, \quad (2)$$

and expected utility is

$$U = \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon}. \quad (3)$$

Let N denote total metropolitan population, fixed in a closed economy. Equilibrium residential population at each location is

$$N_i = N \sum_j \pi_{ij}.$$

Residential floor space market clearing at location i pins down the equilibrium rent:

$$q_i^R = \frac{(1 - \alpha) \sum_j N \pi_{ij} Y_{ij}}{\bar{H}_i^R}. \quad (4)$$

1.3 Firms

At each location j , competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F :

$$Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0, 1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$w_j = \gamma A_j \left(\frac{H_j^F}{L_j} \right)^{1-\gamma}, \quad q_j^F = (1 - \gamma) A_j \left(\frac{L_j}{H_j^F} \right)^\gamma.$$

Commercial floor space is constrained by zoning: $H_j^F = \bar{H}_j^F$. Labor market clearing at j gives

$$L_j = N \sum_i \pi_{ij},$$

determined by household location choices. The disutility $\phi(N_i, L_i)$ uses L_i to denote employment at location i acting as a workplace, which is this same quantity evaluated at $j = i$.

1.4 Municipal Zoning

Land at each location competes for residential and commercial use. Each municipality $g \in G$ chooses an iceberg cost $\{t_i^F\}_{i \in S^g} \geq 0$ on commercial land use at each location: procedural burden — discretionary review, permitting delay, litigation risk, compliance — melts away a fraction of commercial floor space's usable value before it reaches the market, in the spirit of how long approval processes or stringent review requirements shrink the effective commercial return without generating revenue for anyone. Because the same underlying land parcel can be put to either use, competitive landowners allocate it to whichever use pays the higher effective return; if both uses are active in equilibrium, this competition equalizes returns at the margin:

$$q_i^R = \frac{q_i^F}{1 + t_i^F}. \quad (5)$$

Together with the land identity $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$, the residential rent $q_i^R(\bar{H}_i^R)$ from (4) and the commercial rent $q_i^F(\bar{H}_i^F)$ from the firm's first-order condition — each strictly decreasing in its own floor-space argument — condition (5) pins down the equilibrium land split given the iceberg cost. Define

$$g(\bar{H}_i^R; t_i^F) \equiv q_i^R(\bar{H}_i^R) - \frac{q_i^F(\bar{H}_i - \bar{H}_i^R)}{1 + t_i^F}.$$

Since q_i^R is strictly decreasing in $\bar{H}_i^R$ and q_i^F is strictly decreasing in $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ (so $-q_i^F(\bar{H}_i - \bar{H}_i^R)/(1 + t_i^F)$ is itself strictly decreasing in $\bar{H}_i^R$ for any $t_i^F \geq 0$), g is strictly decreasing in $\bar{H}_i^R$, so $g(\bar{H}_i^R; t_i^F) = 0$ has a unique solution $\bar{H}_i^R(t_i^F)$ for every iceberg cost. By the implicit function theorem,

$$\frac{d\bar{H}_i^R}{dt_i^F} = \frac{q_i^F(\bar{H}_i - \bar{H}_i^R)/(1 + t_i^F)^2}{-g'(\bar{H}_i^R)} > 0, \quad g'(\bar{H}_i^R) \equiv \frac{\partial q_i^R}{\partial \bar{H}_i^R} + \frac{1}{1 + t_i^F} \frac{\partial q_i^F}{\partial \bar{H}_i^F} < 0, \quad (6)$$

so a higher iceberg cost on commercial use raises equilibrium residential floor space: melting away more of the commercial return pushes land toward residential use — the same qualitative comparative static as an additive fee, since both the numerator and $-g'(\bar{H}_i^R)$ above are strictly positive. No supplier or developer is introduced — (5) is a no-arbitrage condition on a competitive land market, using only the rent functions the model already has, and total land $\bar{H}_i$ remains fixed.

Unlike an additive fee, the melted portion of commercial value is *definitionally* a real resource loss rather than an assumption: it never becomes usable floor space for anyone, so it is not rebated to residents and does not enter any budget constraint. This keeps household optimization and every downstream object $(\tilde{V}_{ij}, \Phi_i, N_i, q_i^R)$ exactly as derived below, unaffected by the iceberg cost except through (5). The closest empirical analogue remains Glaeser and Gyourko (2002)'s price/construction-cost wedge, read here as a proportional rather than dollar gap: $t_i^F = q_i^F/q_i^R - 1$.

The municipality sets $\{t_i^F\}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government acts as a representative of its residents, who face a tradeoff between lower housing costs — which favor a higher iceberg cost, pushing land toward residential use — and lower congestion and fewer workers near their homes — which favor a lower iceberg cost, leaving land available for commercial use. Formally, municipality g solves

$$\max_{\{t_i^F\}_{i \in S^g}} W^g = \sum_{i \in S^g} N_i \bar{U}_i \quad \text{s.t.} \quad (5), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where N_i is the equilibrium population at location i and $\bar{U}_i = \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon}$ is the expected utility of a representative resident at location i , taking other municipalities' iceberg costs $\{t_{i'}^F\}_{i' \notin S^g}$ and transportation investment $\{I_{kl}\}$ as given. $\phi(N_i, L_i)$ is a disutility from residing at i , capturing two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \quad \frac{\partial \phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined above). I parametrize the disutility in Cobb-Douglas form:

$$\phi(N_i, L_i) = \eta N_i^a L_i^b, \quad \eta > 0, \quad a, b > 0, \quad (7)$$

where η is an overall congestion scale and a, b govern the relative importance of residential crowding versus worker influx. The marginal congestion costs are

$$\frac{\partial \phi}{\partial N_i} = \eta a N_i^{a-1} L_i^b, \quad \frac{\partial \phi}{\partial L_i} = \eta b N_i^a L_i^{b-1}, \quad (8)$$

both positive. Unlike an additively separable disutility, the two congestion sources are here mutual complements: $\partial^2 \phi / \partial N_i \partial L_i = \eta a b N_i^{a-1} L_i^{b-1} > 0$, so an additional resident is more burdensome the more workers already commute into i , and vice versa. Own-argument convexity, $\partial^2 \phi / \partial N_i^2 = \eta a(a-1) N_i^{a-2} L_i^b > 0$, requires $a > 1$, ensuring the first-order condition below has an interior solution.

Characterizing the optimum. Since $\bar{H}_i^R = \bar{H}_i - \bar{H}_i^F(t_i^F)$ is a strictly increasing, invertible function of t_i^F , it is equivalent — and algebraically simpler — to characterize the iceberg-cost choice by differentiating the objective with respect to $\bar{H}_i^R$ directly, then translating back to the iceberg cost at the end. Substituting $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ and differentiating $\sum_{i \in S^g} N_i \bar{U}_i$ with respect to $\bar{H}_i^R$, the interior first-order condition at location $i \in S^g$ is

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (in-movers)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (rent benefit)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial N_i} \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\leq 0 \text{ (congestion cost)}} + N_i \underbrace{\left(-\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial L_i} \frac{\partial L_i}{\partial \bar{H}_i^R} \right)}_{\geq 0 \text{ (less worker influx)}} = 0. \quad (9)$$

The first term is positive: a higher iceberg cost t_i^F pushes land toward residential use, and the resulting looser residential zoning lowers rents, attracting additional residents from the rest of the metro area; each in-mover contributes $\bar{U}_i$ to the municipality's objective. The second term is positive: lower rents raise effective income for all N_i existing residents. The third term is negative: in-movers raise N_i , worsening congestion for all current residents through ϕ . The fourth term is positive: shifting land from commercial to residential reduces employment L_i and the associated worker congestion. The municipality sets the iceberg cost where the benefit of attracting residents and lowering rents is offset by the congestion they impose.

Substituting the Cobb-Douglas marginal congestion costs (8) into the congestion-cost and worker-influx terms makes the interaction between the two channels explicit:

$$\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R} + N_i \frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R} + N_i \frac{\partial \bar{U}_i}{\partial \phi} \eta a N_i^{a-1} L_i^b \frac{\partial N_i}{\partial \bar{H}_i^R} - N_i \frac{\partial \bar{U}_i}{\partial \phi} \eta b N_i^a L_i^{b-1} \frac{\partial L_i}{\partial \bar{H}_i^R} = 0. \quad (10)$$

Because $\partial \bar{U}_i / \partial \phi < 0$ (higher disutility lowers $\tilde{V}_{ij}$, hence Φ_i and $\bar{U}_i$, for every j), the third term is a product of a negative and two non-negative factors and so is ≤ 0 , and the fourth term (after the leading minus sign) is ≥ 0 , matching the signs already established above. The Cobb-Douglas cross-term makes clear that the congestion cost of loosening residential zoning at i is amplified when i already hosts substantial employment L_i ($N_i^{a-1} L_i^b$ rising in L_i for $b > 0$), and symmetrically that the worker-influx benefit of shifting land toward residential use is larger where residential population N_i is already high ($N_i^a L_i^{b-1}$ rising in N_i) — a complementarity the additively separable form could not generate.

Translating back to the iceberg cost. By the chain rule and (6),

$$\frac{dW^g}{dt_i^F} = \frac{dW^g}{d\bar{H}_i^R} \cdot \frac{d\bar{H}_i^R}{dt_i^F}, \quad \frac{d\bar{H}_i^R}{dt_i^F} = \frac{q_i^F(\bar{H}_i - \bar{H}_i^R)/(1 + t_i^F)^2}{-g'(\bar{H}_i^R)} \neq 0, \quad (11)$$

so $dW^g/dt_i^F = 0$ holds if and only if $dW^g/d\bar{H}_i^R = 0$, i.e., if and only if (9) (equivalently (10)) holds at the residential floor space implied by the arbitrage condition (5). Choosing the iceberg cost t_i^F therefore characterizes exactly the same interior solution as the underlying quantity choice; it is simply monotonically rescaled by the strictly positive factor in (11), which does not change where the derivative vanishes.

$\bar{H}_i^F$ itself can equivalently be read as a *quota*: a cap on how much land may be used commercially. The iceberg-cost and quota formulations describe the same one-dimensional municipal choice at i ; the model could equally be stated with a residential-side iceberg cost $t_i^R \geq 0$ defined via $q_i^F = q_i^R/(1 + t_i^R)$, which is strictly decreasing (rather than increasing) in $\bar{H}_i^R$ but satisfies the same equivalence by the symmetric argument. I take t_i^F — an iceberg cost on commercial use — as the primary instrument going forward, since it maps most directly onto how regulatory burden (delay, discretionary review) actually shrinks usable commercial floor space, while remaining recoverable from the same residential and commercial rent data used to measure zoning restrictiveness empirically (Glaeser and Gyourko, 2002).

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{k\ell}\}_{(k,\ell) \in E}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ and the route assignment $\{r_{ij}\}$ as given:

$$\max_{\{I_{k\ell}\}} N \cdot \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{(k,\ell) \in E} \delta_{k\ell} I_{k\ell} \leq K, \quad I_{k\ell} \geq 0,$$

where $\delta_{k\ell}$ is the per-unit cost of investment on edge (k, ℓ) and K is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. Because $\ln \tau_{ij} = \sum_{(k,\ell) \in r_{ij}} \ln d_{k\ell}$, investment on edge (k, ℓ) only affects the commuting cost of a residence-workplace pair (i, j) whose route uses that edge, through $\partial \ln \tau_{ij} / \partial I_{k\ell} = -\gamma / I_{k\ell}$ for $(k, \ell) \in r_{ij}$ (and 0 otherwise). Aggregating over all such pairs, the interior first-order condition for edge (k, ℓ) is

$$\frac{N\gamma}{I_{k\ell}} \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon-1} \sum_{(i,j) : (k,\ell) \in r_{ij}} T_i E_j \tilde{V}_{ij}^\epsilon = \mu \delta_{k\ell}. \quad (12)$$

The left-hand side is the marginal social benefit of investment on edge (k, ℓ) : lower travel cost on that edge raises $\tilde{V}_{ij}$, and hence Φ , for *every* residence-workplace pair routed through it — not only pairs directly adjacent to k and ℓ . This is the sense in which investment spillovers arise endogenously from the network structure: an edge shared by many routes is more valuable to the authority than one used only by its own endpoints, holding the direct commuting-cost elasticity fixed. The authority equates this marginal benefit per dollar of investment across all active edges.

1.6 Equilibrium

Definition 1 (Decentralized Equilibrium). A decentralized equilibrium is a profile

$$\left(\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}_{i \in S}, \{I_{k\ell}^*\}_{(k,\ell) \in E}, \{q_i^{R*}, q_i^{F*}, w_j^*\}_{i,j \in S} \right)$$

such that:

1. **Household optimality.** Commuting flows $\{N\pi_{ij}^*\}$ are given by the Fréchet probability formula at prices (q_i^{R*}, w_j^*) and investment I^* , with route-based commuting costs $\{\tau_{ij}^*\}$ and edge traffic $\{n_{k\ell}^*\}$ satisfying (1).
2. **Firm optimality.** Wages w_j^* and commercial rents q_j^{F*} satisfy the firm first-order conditions at $H_j^F = \bar{H}_j^{F*}$ and $L_j^* = N \sum_i \pi_{ij}^*$.
3. **Market clearing.** Residential floor space clears at q_i^{R*} given by (4), commercial floor space is fixed at $H_j^F = \bar{H}_j^{F*}$, and the labor market clears at $L_j^* = N \sum_i \pi_{ij}^*$.
4. **Municipal Nash equilibrium.** Each municipality g sets the iceberg cost $\{t_i^{F*}\}_{i \in S^g}$ — equivalently, induces $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}_{i \in S^g}$ via (11) — to satisfy the first-order condition (9), given $\{\bar{H}_{i'}^{R*}, \bar{H}_{i'}^{F*}\}_{i' \notin S^g}$ and I^* ; no municipality has a profitable deviation.
5. **Transportation authority optimality.** $\{I_{k\ell}^*\}$ satisfies the first-order condition (12) for every edge $(k, \ell) \in E$, maximizing aggregate welfare given $\{\bar{H}_i^{R*}, \bar{H}_i^{F*}\}$ subject to the budget constraint.

Conditions (iv) and (v) jointly constitute the simultaneous Nash equilibrium between municipalities and the transportation authority: each set of agents takes the other's choices as given and best-responds. The equilibrium iceberg cost $t_i^{F*} \equiv q_i^{F*}/q_i^{R*} - 1$ is computed from the equilibrium quantities in this profile, not solved for as a separate object.

1.7 Quantification

The household block delivers a log-linear estimating system once the congestion term is handled: writing $w_j - \phi_i = w_j(1 - \phi_i/w_j)$ and taking a first-order log approximation, $\ln(w_j - \phi_i) \approx \ln w_j - \phi_i/w_j$. Substituting into $\ln \pi_{ij} = \ln T_i + \ln E_j + \epsilon \ln \bar{V}_{ij} - \ln \Phi$ with $\tau_{ij} = \prod_{(k,\ell) \in r_{ij}} \delta_{k\ell}^\tau n_{k\ell}^\rho / I_{k\ell}^\gamma$ gives a structural-gravity regression on observed commuting flows $N\pi_{ij}$:

$$\begin{aligned} \ln \pi_{ij} \approx & \underbrace{\left[\ln T_i + \epsilon \ln B_i - \epsilon(1 - \alpha) \ln q_i^R \right]}_{\text{origin FE}} + \underbrace{\left[\ln E_j + \epsilon \ln w_j \right]}_{\text{destination FE}} - \epsilon \frac{\phi_i}{w_j} \\ & - \epsilon \sum_{(k,\ell) \in r_{ij}} \ln \delta_{k\ell}^\tau - \epsilon \rho \sum_{(k,\ell) \in r_{ij}} \ln n_{k\ell} + \epsilon \gamma \sum_{(k,\ell) \in r_{ij}} \ln I_{k\ell} - \ln \Phi, \end{aligned} \quad (13)$$

in the spirit of Ahlfeldt et al. (2015). The regressors $\ln n_{ij}, \ln I_{ij}$ of the OD-level friction become route-level sums $\sum_{(k,\ell) \in r_{ij}} \ln n_{k\ell}, \sum_{(k,\ell) \in r_{ij}} \ln I_{k\ell}$ once routes span multiple edges — a measurement change (route-level GIS data replace a single OD-level number) rather than an identification problem, since the underlying elasticities ρ, γ are still edge-level primitives. With origin and destination fixed effects, the coefficients on these route sums and on ϕ_i/w_j identify $-\epsilon\rho, \epsilon\gamma$, and $-\epsilon$ respectively. Given ϵ , the origin fixed effect nets out $\ln T_i + \epsilon \ln B_i$ as a residual (observed q_i^R, α), the standard model-inversion step.

The zoning iceberg cost requires no approximation: (5) gives $t_i^F = q_i^F/q_i^R - 1$ directly from residential and commercial rent data, with no estimation. Given a constructed $\hat{\phi}_i$ from the step above, the congestion parameters follow from

$$\ln \hat{\phi}_i = \ln \eta + a \ln N_i + b \ln L_i + u_i, \quad (14)$$

recovering (η, a, b) directly, with $a > 1$ testable against the convexity restriction.

The comparative statics inside (10) ($\partial N_i/\partial \bar{H}_i^R$ etc.) are a further object: N_i, q_i^R, w_i, ϕ_i are jointly determined through $\Phi = \sum_{i'} \Phi_{i'}$ and the labor market, so their response to a zoning change is a general-equilibrium comparative static, not a single-equation partial derivative — exactly why counterfactuals in this literature are solved as a linearized system rather than a closed-form elasticity. A tractable special case treats each location as small relative to the metro ($d \ln \Phi \approx 0$, wages elsewhere unaffected), collapsing the metro-wide fixed point to a local system in $(\hat{N}_i, \hat{L}_i, \hat{q}_i^R, \hat{w}_i, \hat{\phi}_i)$ solvable by ordinary matrix inversion; working this out is the natural next step for the quantitative exercise.

1.8 Data and Calibration

Following Fajgelbaum and Schaal (2020)’s treatment of network and cost fundamentals, parameters split into three groups: estimated from the model’s own equations (§Quantification), set from external estimates, and location/link fundamentals recovered as residuals or calibrated directly to observed infrastructure costs.

Data. Commuting flows $N\pi_{ij}$ and employment L_i : LODES/LEHD origin-destination data. Wages w_j and population N_i : ACS. Residential and commercial rents q_i^R, q_i^F : county appraisal district records or commercial listing data (CoStar), jointly identifying t_i^F via (5). Land area $\bar{H}_i$: parcel/GIS data. Network edges E , route assignment $\{r_{ij}\}$, and travel times, for $\delta_{k\ell}^\tau$: GIS road network shortest paths and observed transit line/transfer sequences. Historical investment $I_{k\ell}$ and per-unit construction costs $\delta_{k\ell}$: NTD and state DOT capital programs, at the edge level. Total budget K : observed metro-area transportation capital spending over the relevant period.

Estimated from model equations. ϵ, ρ , and the commuting-cost investment elasticity in (13); (η, a, b) from the congestion regression; t_i^F is not estimated at all, but read directly off rent data via (5).

Set externally. α , the consumption expenditure share, from the CEX or national accounts; γ , the firm’s labor share in (4)’s underlying production function, from the local labor share of income (BEA). *Notation note:* the commuting-cost investment elasticity in $d_{k\ell} = \delta_{k\ell}^\tau n_{k\ell}^\rho / I_{k\ell}^\gamma$ currently reuses the symbol γ already assigned to the firm’s labor share in $Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma}$ — these are distinct parameters with distinct empirical targets and should be given separate symbols before the calibration is finalized.

Fundamentals recovered as residuals. T_i, B_i (jointly, from the origin fixed effect) and E_j, A_j (from the destination fixed effect and the firm’s first-order condition) so that the model exactly matches observed N_i, L_i, w_j, q_i^R at the calibrated parameters, following the model-inversion logic of Ahlfeldt et al. (2015) and Fajgelbaum and Schaal (2020).

References

- [1] Gabriel M Ahlfeldt et al. “The economics of density: Evidence from the Berlin wall”. en. In: *Econometrica: journal of the Econometric Society* 83 (6 2015), pp. 2127–2189.
- [2] Treb Allen and Costas Arkolakis. “The Welfare Effects of Transportation Infrastructure Improvements”. en. In: *The Review of Economic Studies* 89 (6 Nov. 7, 2022), pp. 2911–2957.
- [3] Olivia Bordeu. *Commuting Infrastructure in Fragmented Cities*. 2026.
- [4] Pablo D Fajgelbaum and Edouard Schaal. “Optimal transport networks in spatial equilibrium”. en. In: *Econometrica: Journal of the Econometric Society* 88 (4 2020), pp. 1411–1452.
- [5] Edward L Glaeser and Joseph Gyourko. *The Impact of Zoning on Housing Affordability*. en. Research rep. 8835. Cambridge, MA, 2002.
- [6] Evan Mast. “Warding off Development: Local Control, Housing Supply, and NIM-BYs”. en. In: *The Review of Economics and Statistics* 106 (3 May 14, 2024), pp. 671–680.
- [7] Vincent Rollet and Laura Weiwu. *Measuring Winners and Losers from Increasing Housing Supply*. en. Research rep. 2025.